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Fall 2024 MAT 2362
Instructor: Kirill Zaynullin

Final exam

First name: _____ Last name: _____

Student number:

Please, read the following instructions carefully:

- This is a **closed book exam**. All electronic communication devices (e.g. cell phones, calculators, laptops, tablets) must be turned off and **put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device (even if it is switched off), academic fraud allegations will be filed.
- Fill in your first name, the last name and the student number. **Do not detach** the pages of this examination.

This space is reserved for marker

Nr	1	2	3	4	5	6	7	8	9	Total
Mark										
Out of	2	3	2	4	3	2	4	4	3	27

1. Using truth tables, determine whether the following formula is a tautology, a contradiction, or none `(2pt)`

$$(X \Rightarrow Y) \vee (Y \Rightarrow X)$$

(correct answer without justifications is 0pts)

Proof:

2. Define a relation $\sim$ on $\mathbb{R}^2$ by stating that $(a, b) \sim (c, d)$ if and only if $a^4 + b^4 \leq c^4 + d^4$.
Prove that $\sim$ is reflexive and transitive but not symmetric. (3pt)

(when showing that it is not symmetric, provide an explicit numerical counterexample)

Proof:

3. Prove directly that if n is an odd integer, then $n^2 - 1$ is divisible by 4. (2pt)

Proof:

4. (a) Write down the definition of a minimal and of a maximal element in a partially ordered set. (1pt)

(b) Let $(X, \preceq)$ be a totally ordered set.
Prove that if x_o is the **minimal** element, then it is also a **minimum** element. (3pt)

Proof:

5. Prove that $Func(\mathbb{N}, \mathbb{N}) \sim \mathfrak{p}(\mathbb{N})$. (3pt)

Hint: use the graph of a function to define an injection into the power set $\mathfrak{p}(\mathbb{N} \times \mathbb{N})$

Proof:

6. Prove using the axioms of set theory that $\{\{a\},\{a,b\}\} = \{\{b'\},\{a',b'\}\}$ iff $a = a'$ and $b = b'$ (2pt)

In the proof you must explicitly state which axioms you are using.

Proof:

7. Prove using the **ordinal induction** that if α and β are ordinals, then $\beta \leq \alpha + \beta$. (4pt)

Hint: You must prove the base step, the successor step and the limit step of the induction.

Proof:

8. Suppose $X = \{0, 1, 2, \ldots\}$ is the usual well-ordered set of non-negative integers. Consider another well-ordered set

$$Y = X \times \{0, 1\} \text{ with } (x, i) < (x', j) \text{ iff } i < j \text{ or } i = j \text{ and } x < x'.$$

Prove that there is no bijection between X and Y that preserves the well-orderings. (4pt)

(Hint: Identify sets with ordinals and use comparison results for ordinals)

Proof:

9. Consider the set $\mathbb{R}$ endowed with the following binary operation $\star$:

$$x \star y = x + y - 1.$$

Determine whether this operation is associative, commutative, or it has an identity. Justify your answers. `(3pt)`

Correct answers without justification is 0pts

Proof:

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